

CODE:

```
from sklearn.tree import DecisionTreeClassifier, _tree
```

```
data = [
```

```
    ['Red', 'Round', 'Medium', 'Apple'],
```

```
    ['Green', 'Round', 'Medium', 'Apple'],
```

```
    ['Red', 'Round', 'Large', 'Apple'],
```

```
    ['Green', 'Round', 'Large', 'Apple'],
```

```
    ['Yellow', 'Long', 'Medium', 'Banana'],
```

```
    ['Yellow', 'Long', 'Large', 'Banana'],
```

```
    ['Yellow', 'Long', 'Small', 'Banana'],
```

```
    ['Orange', 'Round', 'Medium', 'Orange'],
```

```
    ['Orange', 'Round', 'Large', 'Orange'],
```

```
    ['Orange', 'Round', 'Small', 'Orange']
```

```
]
```

```
color = {'Red': 1, 'Green': 2, 'Yellow': 3, 'Orange': 4}
```

```
shape = {'Round': 1, 'Long': 2}
```

```
size = {'Small': 1, 'Medium': 2, 'Large': 3}
```

```
feature_names = ['Color', 'Shape', 'Size']
```

```
X = []
```

```
for row in data:
```

```
    X.append([
```

```
        color[row[0]],
```

```
        shape[row[1]],
```

```
        size[row[2]]
```

```
    ])
```

```
y = [row[3] for row in data]
```

```
model = DecisionTreeClassifier(criterion="entropy")
```

```
model.fit(X, y)
```

```

tree = model.tree_

print("\nDECISION TREE STRUCTURE\n")

def print_tree(node, indent=""):
    if tree.feature[node] != _tree.TREE_UNDEFINED:
        feature = feature_names[tree.feature[node]]
        threshold = tree.threshold[node]

        print(indent + "Question: " + feature + " <= " + str(round(threshold, 2)))
        print(indent + "|-- YES")
        print_tree(tree.children_left[node], indent + "| ")
        print(indent + "|-- NO")
        print_tree(tree.children_right[node], indent + "| ")
    else:
        prediction = model.classes_[tree.value[node][0].argmax()]
        print(indent + "Prediction: " + prediction)

print_tree(0)

print("\nEnter new fruit details")

new_color = input("Color (Red/Green/Yellow/Orange): ")
new_shape = input("Shape (Round/Long): ")
new_size = input("Size (Small/Medium/Large): ")

new_fruit = [[
    color[new_color],
    shape[new_shape],
    size[new_size]
]]

prediction = model.predict(new_fruit)

print("\nPredicted Fruit:", prediction[0])

```

DECISION TREE STRUCTURE

Question: Color ≤ 2.5

|-- YES

| Prediction: Apple

|-- NO

| Question: Shape ≤ 1.5

| |-- YES

| | Prediction: Orange

| |-- NO

| | Prediction: Banana

Enter new fruit details

Predicted Fruit: Apple